Using trees to grow money

Hedge fund investors want reassurance that their hedge funds are following the strategies they say they are following. Maria Augusta Miceli and Gabriele Susinno show how the analysis of complex systems in physics, engineering and biology can be used to determine how assets are growing

High returns, low volatility and low correlation to general behaviour in the market. These are the hedge fund investment qualities that appeal most to both private and institutional buyers. The challenge to those investors is that hedge fund managers enjoy a large degree of freedom to exercise proprietary investment strategies, and to use advanced technologies to exercise those strategies. These factors limit transparency, and often lead to a difference between stated and actual hedge fund investment strategies (see Lo, 2001). The managers of funds of hedge funds are particularly concerned about 'style creep': the tendency of individual hedge funds to depart from their stated investment strategy when they see better returns from other strategies.

We have seen recent moves by regulators to force hedge funds to issue regular statements about their investment strategy, but the fact remains that traditional due diligence practices will fail to disclose the hedge fund manager's policy in this respect, so hedge fund investors must be ready to sign a contract of trust.

How can we improve on that when it comes to testing the true source of returns? Since each declared investment strategy may in fact originate from a multiplicity of management decisions, some questions might arise as a matter of course:

- ☐ Are realised returns consistent with the 'declared' strategy?
- ☐ How can the realised returns of hedge funds be monitored and compared with the results of other hedge funds belonging to the same category of declared strategy?
- ☐ Is it possible to quantify the strategies to optimise the selection in funds of funds diversification?

We answer these questions by statistical methods. The most common approach used to define the degree of similarity among returns is the correlation coefficient. However, there is another possible alternative that lies in the combinatorial optimising techniques useful to automatically classify complex data structures. These methods are commonly used in physics, IT or biology to classify genomes (Xu, Olman & Xu, 2001, Genome Research, 2003, and Rammal, Toulouse & Virasoro, 1986), but they are rarely applied in finance (Mantegna, 1999).

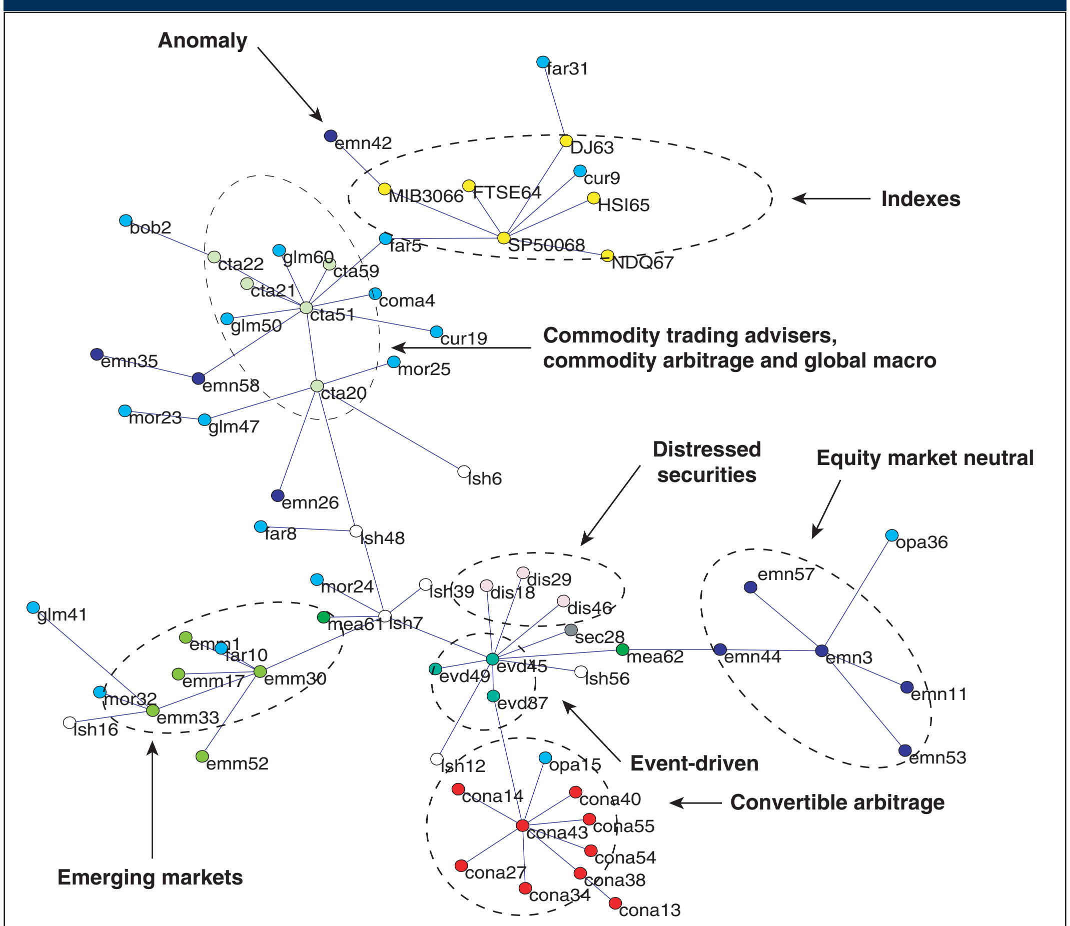
When applied to hedge fund return time series, these techniques allow us to determine and to visualise a taxonomy embedded in synchronous historical data. It becomes possible to classify funds in clusters by maximising homogeneity within groups of funds and by maximising heterogeneity between groups of funds.

Underlying strategies

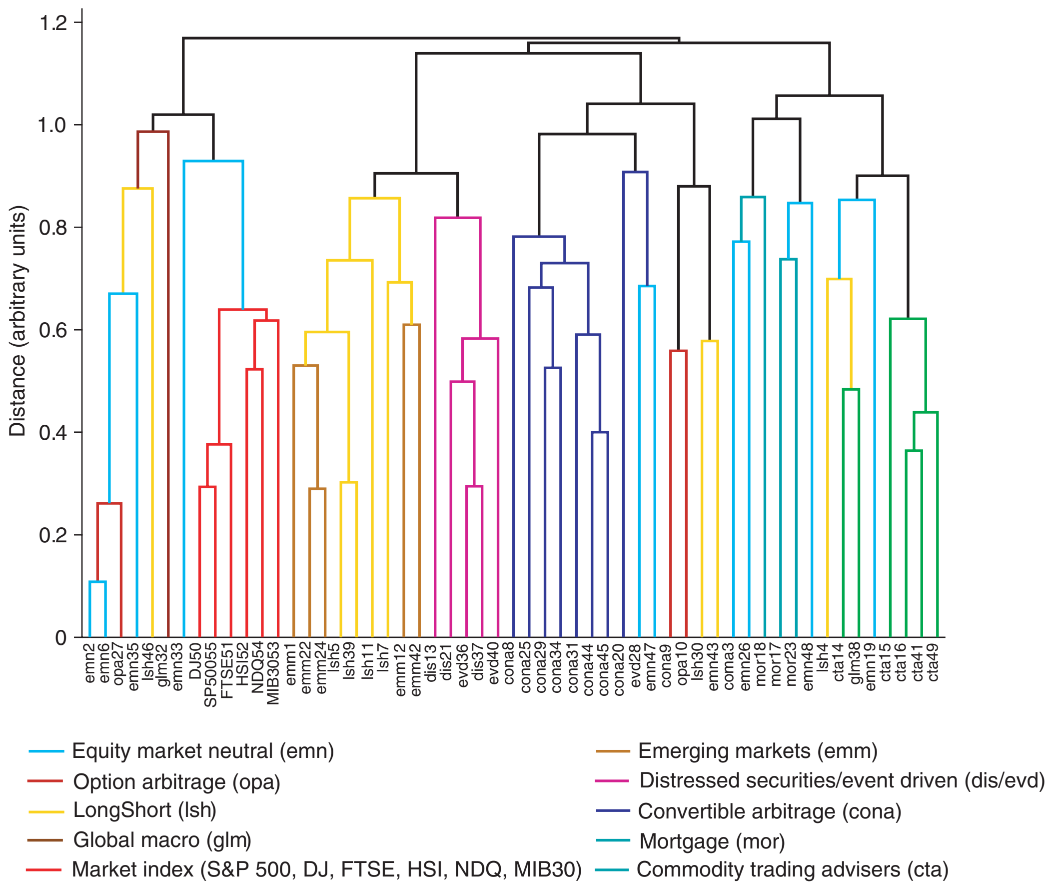
By analysing those characteristics that cluster funds together and those that diversify them, it is possible to recognise the real underlying strategies and, correspondingly, to find managers that do not follow them. In this article, we briefly introduce the combinatorial optimising method, then concentrate on its use in the process of portfolio selection and diversification. Technical details are available in Miceli & Susinno (2003).

The method is based on a similarity concept expressed by distance: similarity grows as distance shrinks. A frequently used measure is the Euclidean distance between two time series, which can be calculated starting from the linear correlation coefficient (Mantegna & Stanley, 2000). Given the distances between all funds, and by considering each fund as a node in a tree structure, it is possible to determine a data representation known as a 'minimum spanning tree' (MST). This is the unique

1. Minimum spanning tree



2. Dendrogram representation and strategy definitions



graph that connects all nodes with the minimum extension. The minimum spanning tree is obtainable through simple algorithms (see Papadimitriou & Steiglitz, 1998).

Distance minimisation between tree nodes allows for a natural classification of hedge funds in clusters containing elements with similar realisations. The further merger of smaller clusters in larger clusters leads to a hierarchical classification – called a dendrogram – that shows relations between clusters. Arresting that merger process at a desired level gives us a clustering of disjointed groups.

Example

In our example, using a return time series of 62 hedge funds[1] over the period January 2000 to January 2003, we determined the associated MST (figure 1) and dendrogram (figure 2).[2] We note how the clustering process leads to a meaningful economic classification of strategies, and how anomalies can be promptly detected.

For some strategies, the cluster is well defined and there exists a fund that acts as a reference centre (cona 43, cta51, emm30, etc). For other strategies, such as 'longshort', there is not a well-defined cluster. This could mean that there are no peculiar characteristics in this management style. This observation may suggest that the less discretionary the hedge fund strategies or management policies are, the more similar their corresponding returns. Examples of less discretionary hedge fund strategies are those that follow mathematical models implemented by software decisions.

In a universe with low transparency from an investor's point of view, and where operating strategies are protected from full disclosure, data mining and the instruments of classification we have described would allow the investor to extract the maximum information from available data, and to verify the declared strategies in the statements issued by hedge funds.

In particular, nodes anomalous to a reference cluster help to identify funds that require a deeper investigation. Beyond qualitative control, these results are useful in maximising portfolio diversification. The analyst can select funds belonging to different clusters, and pay special attention to those characterising central nodes. Moreover, those characteristics that identify a cluster help to define the actual benchmark for a given strategy.

Finally, the tree structure offers an objective basis for extracting economic conclusions, selecting an economic portfolio then controlling that portfolio. By extracting the structure hidden in large correlation matrices, trees are easier to interpret than those large correlation matrices in their raw form. □

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[1] Funds' names are omitted to protect managers' privacy
[2] Figure 1 was created from an heterogeneous set of hedge fund strategies. Strategy acronyms are defined in figure 2

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